Problem

There are two states, Red and Blue, initially with 100 residents each. Every resident supports exactly one of two parties, Democrat or Republican. Initially, 60 residents of Blue and 30 residents of Red are Democrats.

Ten residents move from Blue to Red, and nobody changes parties. Let k be the number of Democrats among the ten people who move.

Whenever a resident is selected, the selection is uniform over all 200 people. Let D be the event that the selected resident is a Democrat, and let B be the event that the resident lives in Blue at the time of selection.

(a) Find $P(D \mid B)$, $P(D \mid B^c)$, and $P(D)$ before the move. Does the move change $P(D)$? Explain.

(b) Determine all integer values of k for which the proportion of Democrats increases strictly in both states.

Explain, using the law of total probability with the appropriate weights, how both proportions can increase while the overall proportion stays unchanged.

(c) For this part, take $k = 5$ and consider the population after the move. A survey interviews the selected resident only if their name appears on a list containing all Blue residents and all Democrats in Red.

Let A be the event that the selected resident is on this list. Find $P(D \mid A)$ using the law of total probability conditional on A, with the partition $\{B, B^c\}$, and state the two group weights.

Explain why this probability differs from the proportion of Democrats in the whole population.

Solution

(a)

Before the move,

$$\because \quad P(B) = P(B^c) = \frac{100}{200} = \frac{1}{2}$$

$$P(D \cap B) = \frac{60}{200}, \qquad P(D \cap B^c) = \frac{30}{200}.$$

By conditional probability,

$$\begin{aligned} P(D \mid B) &= \frac{P(D \cap B)}{P(B)} \\ &= \frac{60/200}{100/200} \\ &= \frac{3}{5} \end{aligned}$$

Similarly,

$$\begin{aligned} P(D \mid B^c) &= \frac{P(D \cap B^c)}{P(B^c)} \\ &= \frac{30/200}{100/200} \\ &= \frac{3}{10} \end{aligned}$$

By the law of total probability,

$$\begin{aligned} P(D) &= P(D \mid B)P(B) + P(D \mid B^c)P(B^c) \\ &= \frac{3}{5} \cdot \frac{1}{2} + \frac{3}{10} \cdot \frac{1}{2} \\ &= \frac{9}{20} \end{aligned}$$

After the move,

$$\begin{aligned} P'(B) &= \frac{90}{200}, \\ P'(B^c) &= \frac{110}{200}, \\ P'(D \mid B) &= \frac{60-k}{90}, \\ P'(D \mid B^c) &= \frac{30+k}{110}. \end{aligned}$$

$$\begin{aligned} \therefore \quad P'(D) &= P'(D \mid B)P'(B) + P'(D \mid B^c)P'(B^c) \\ &= \frac{60-k}{90} \cdot \frac{90}{200} + \frac{30+k}{110} \cdot \frac{110}{200} \\ &= \frac{60-k+30+k}{200} \\ &= \frac{90}{200} \\ &= \frac{9}{20}. \end{aligned}$$

$$\implies \quad P'(D) = P(D) = \frac{9}{20}$$

The move changes neither the total population nor the total number of Democrats.

(b)

After the move, the Democratic proportions are

$$P'(D \mid B) = \frac{60-k}{90}, \qquad P'(D \mid B^c) = \frac{30+k}{110}.$$

For Blue,

$$\begin{aligned} \frac{60-k}{90} &> \frac{60}{100} \\ 60-k &> 54 \\ k &< 6. \end{aligned}$$

For Red,

$$\begin{aligned}\frac{30+k}{110} &> \frac{30}{100}\\ 30+k &> 33\\ k &> 3.\end{aligned}$$

$$\because \quad 3 < k < 6, \qquad k \in \mathbb{Z}$$

$$\implies \quad k \in \{4, 5\}$$

Before the move, the population weights are

$$P(B) = \frac{1}{2}, \qquad P(B^c) = \frac{1}{2}.$$

$$\begin{aligned}\therefore \quad P(D) &= \frac{3}{5} \cdot \frac{1}{2} + \frac{3}{10} \cdot \frac{1}{2}\\ &= \frac{9}{20}.\end{aligned}$$

After the move, the population weights become

$$P'(B) = \frac{9}{20}, \qquad P'(B^c) = \frac{11}{20}.$$

By the law of total probability,

$$\begin{aligned}P'(D) &= P'(D \mid B)P'(B) + P'(D \mid B^c)P'(B^c)\\ &= \frac{60-k}{90} \cdot \frac{9}{20} + \frac{30+k}{110} \cdot \frac{11}{20}\\ &= \frac{60-k}{200} + \frac{30+k}{200}\\ &= \frac{9}{20}.\end{aligned}$$

For example, when $k = 5$,

$$\begin{aligned}P'(D \mid B) &= \frac{55}{90} > \frac{60}{100},\\ P'(D \mid B^c) &= \frac{35}{110} > \frac{30}{100}.\end{aligned}$$

However,

$$\begin{aligned}P'(D) &= \frac{55}{90} \cdot \frac{90}{200} + \frac{35}{110} \cdot \frac{110}{200}\\ &= \frac{55+35}{200}\\ &= \frac{9}{20}.\end{aligned}$$

$$\implies \quad P'(D) = P(D) = 0.45$$

Both conditional proportions increase, but their population weights change from $(0.5, 0.5)$ to $(0.45, 0.55)$.

The larger weight is assigned to the state with the lower Democratic proportion. The change in weights offsets the increases in the conditional proportions.

(c)

$$\because \quad k = 5$$

After the move,

	Blue	Red
Population	90	110
Democrats	55	35
Republicans	35	75

The list contains all 90 Blue residents and the 35 Democrats in Red.

$$\therefore \quad |A| = 90 + 35 = 125.$$

The two group weights conditional on A are

$$P(B \mid A) = \frac{90}{125} = \frac{18}{25}$$
$$P(B^c \mid A) = \frac{35}{125} = \frac{7}{25}$$

For the Blue group,

$$\because \quad A \cap B = B$$

$$\therefore \quad P(D \mid A \cap B) = \frac{55}{90} = \frac{11}{18}.$$

For the Red group, every resident on the list is a Democrat.

$$\therefore \quad P(D \mid A \cap B^c) = 1.$$

By the conditional law of total probability,

$$\begin{aligned} P(D \mid A) &= P(D \mid A \cap B)P(B \mid A) \\ &\quad + P(D \mid A \cap B^c)P(B^c \mid A) \\ &= \frac{55}{90} \cdot \frac{90}{125} + 1 \cdot \frac{35}{125} \\ &= \frac{55 + 35}{125} \\ &= \frac{90}{125}. \end{aligned}$$

$$\implies \quad P(D \mid A) = \frac{18}{25} = 0.72$$

For the whole population,

$$P(D) = \frac{90}{200} = 0.45.$$

$$\because \quad |D \cap A| = 90, \qquad |A| = 125, \qquad |\Omega| = 200$$

$$\therefore \quad P(D \mid A) = \frac{90}{125} > \frac{90}{200} = P(D).$$

The list excludes all 75 Republicans in Red but includes every Democrat.

Thus, the selected group overrepresents Democrats relative to the whole population.

$$\implies \quad P(D \mid A) = 0.72 > 0.45 = P(D)$$

Problem 2

A sequence of $n \geq 1$ mutually independent trials is performed. Each trial results in either success or failure, and the probability of success on trial i is p_i, where $0 \leq p_i \leq 1$. Let A_n be the event that the number of successes among the first n trials is even.

(a) Find $P(A_2)$ by listing the possible outcomes, and show that

$$P(A_2) = \frac{1 + (1 - 2p_1)(1 - 2p_2)}{2}.$$

(b) By considering whether the last trial succeeds or fails, obtain a recursion for $P(A_n)$ for $n \geq 2$. Use the recursion to prove that

$$P(A_n) = \frac{1 + \prod_{i=1}^{n}(1 - 2p_i)}{2}.$$

You may use mathematical induction or repeated substitution in the recursion. State the initial value explicitly.

(c) Under the independence assumption, is $P(A_n) = 1/2$ if and only if at least one $p_i = 1/2$? Justify your answer. Then remove the independence assumption and specify probabilities for the four possible outcomes of two trials so that $p_1 = p_2 = 1/2$ but $P(A_2) \neq 1/2$.

Solution

(a)

Let S and F denote success and failure, respectively.

The four possible outcomes are

$$\{SS, SF, FS, FF\}.$$

$$\begin{aligned} P(SS) &= p_1 p_2, \\ P(SF) &= p_1(1 - p_2), \\ P(FS) &= (1 - p_1)p_2, \\ P(FF) &= (1 - p_1)(1 - p_2). \end{aligned}$$

$$A_2 = \{SS, FF\}.$$

$$\begin{aligned}
\therefore \quad P(A_2) &= P(SS) + P(FF) \\
&= p_1 p_2 + (1 - p_1)(1 - p_2) \\
&= 1 - p_1 - p_2 + 2p_1 p_2 \\
&= \frac{2 - 2p_1 - 2p_2 + 4p_1 p_2}{2} \\
&= \frac{1 + (1 - 2p_1)(1 - 2p_2)}{2}
\end{aligned}$$

(b)

Let S_n and F_n denote success and failure on the n-th trial.

$$A_n = (A_{n-1} \cap F_n) \cup (A_{n-1}^c \cap S_n).$$

$$\begin{aligned}
\therefore \quad P(A_n) &= P(A_{n-1} \cap F_n) + P(A_{n-1}^c \cap S_n) \\
&= P(A_{n-1})(1 - p_n) + [1 - P(A_{n-1})]p_n \\
&= (1 - 2p_n)P(A_{n-1}) + p_n.
\end{aligned}$$

Initial value:

$$P(A_1) = 1 - p_1.$$

Repeated substitution:

$$\begin{aligned}
P(A_n) - \frac{1}{2} &= (1 - 2p_n)P(A_{n-1}) + p_n - \frac{1}{2} \\
&= (1 - 2p_n)\left(P(A_{n-1}) - \frac{1}{2}\right).
\end{aligned}$$

$$\begin{aligned}
\therefore \quad P(A_n) - \frac{1}{2} &= (1 - 2p_n)\left(P(A_{n-1}) - \frac{1}{2}\right) \\
&= (1 - 2p_n)(1 - 2p_{n-1})\left(P(A_{n-2}) - \frac{1}{2}\right) \\
&= \cdots \\
&= \prod_{i=2}^{n}(1 - 2p_i)\left(P(A_1) - \frac{1}{2}\right) \\
&= \prod_{i=2}^{n}(1 - 2p_i)\left(\frac{1}{2} - p_1\right) \\
&= \frac{1}{2}\prod_{i=2}^{n}(1 - 2p_i)(1 - 2p_1) \\
&= \frac{1}{2}\prod_{i=1}^{n}(1 - 2p_i).
\end{aligned}$$

$$\Rightarrow \quad P(A_n) = \frac{1 + \prod_{i=1}^{n}(1 - 2p_i)}{2}$$

(c)

Under independence:

$$P(A_n) = \frac{1 + \prod_{i=1}^{n}(1 - 2p_i)}{2}.$$

$$\begin{aligned} P(A_n) = \frac{1}{2} &\iff 1 + \prod_{i=1}^{n}(1 - 2p_i) = 1 \\ &\iff \prod_{i=1}^{n}(1 - 2p_i) = 0 \\ &\iff \exists\, i \in \{1, \ldots, n\}, \quad 1 - 2p_i = 0 \\ &\iff \exists\, i \in \{1, \ldots, n\}, \quad p_i = \frac{1}{2}. \end{aligned}$$

$$\therefore \quad P(A_n) = \frac{1}{2} \iff \exists\, i, \quad p_i = \frac{1}{2}$$

Without independence:

Consider the joint distribution

$$\begin{aligned} P(SS) &= \frac{1}{2}, \\ P(SF) &= 0, \\ P(FS) &= 0, \\ P(FF) &= \frac{1}{2}. \end{aligned}$$

$$\begin{aligned} p_1 &= P(SS) + P(SF) \\ &= \frac{1}{2} + 0 = \frac{1}{2}, \end{aligned}$$

$$\begin{aligned} p_2 &= P(SS) + P(FS) \\ &= \frac{1}{2} + 0 = \frac{1}{2}. \end{aligned}$$

$$\begin{aligned} P(A_2) &= P(SS) + P(FF) \\ &= \frac{1}{2} + \frac{1}{2} \\ &= 1 \neq \frac{1}{2}. \end{aligned}$$

$$\because \quad P(SS) = \frac{1}{2} \neq p_1 p_2 = \frac{1}{4},$$

$$\therefore \quad \text{the two trials are not independent.}$$

$$p_1 = p_2 = \frac{1}{2}, \qquad P(A_2) = 1 \neq \frac{1}{2}.$$

Problem 3

To battle against spam, Bob installs two anti-spam programs. An email is either legitimate (event L) or spam (event L^c). Let M_j be the event that program j marks the email as legitimate, for $j = 1, 2$. Assume that

$$P(L) = 0.10, \qquad P(M_j \mid L) = P(M_j^c \mid L^c) = 0.90 \quad (j = 1, 2).$$

The two programs process the same email. Their outputs are conditionally independent given L, and also conditionally independent given L^c.

(a) Find $P(L \mid M_1)$ and simplify your answer.

(b) Find $P(L \mid M_1 \cap M_2)$ and $P(L \mid M_1 \cap M_2^c)$. Compare each with the prior probability $P(L)$.

(c) Are M_1 and M_2 independent without conditioning on the email's true status? Prove your answer by computing the required probabilities. Explain why your answer is consistent with the stated conditional independence.

Solution

(a)

$$\because \quad P(L^c) = 0.90, \qquad P(M_1 \mid L^c) = 0.10.$$

By the law of total probability,

$$\begin{aligned} P(M_1) &= P(M_1 \mid L)P(L) + P(M_1 \mid L^c)P(L^c) \\ &= 0.90(0.10) + 0.10(0.90) \\ &= 0.18. \end{aligned}$$

By Bayes' theorem,

$$\begin{aligned} P(L \mid M_1) &= \frac{P(M_1 \mid L)P(L)}{P(M_1)} \\ &= \frac{0.90(0.10)}{0.18} \\ &= \frac{1}{2}. \end{aligned}$$

$$\therefore \quad P(L \mid M_1) = 0.50.$$

(b)

Case 1: $P(L \mid M_1 \cap M_2)$.

By conditional independence,

$$\begin{aligned} P(M_1 \cap M_2 \mid L) &= P(M_1 \mid L)P(M_2 \mid L) \\ &= 0.90^2 = 0.81, \\ P(M_1 \cap M_2 \mid L^c) &= P(M_1 \mid L^c)P(M_2 \mid L^c) \\ &= 0.10^2 = 0.01. \end{aligned}$$

$$\begin{aligned} P(M_1 \cap M_2) &= P(M_1 \cap M_2 \mid L)P(L) \\ &\quad + P(M_1 \cap M_2 \mid L^c)P(L^c) \\ &= 0.81(0.10) + 0.01(0.90) \\ &= 0.09. \end{aligned}$$

By Bayes' theorem,

$$\begin{aligned} P(L \mid M_1 \cap M_2) &= \frac{P(M_1 \cap M_2 \mid L)P(L)}{P(M_1 \cap M_2)} \\ &= \frac{0.81(0.10)}{0.09} \\ &= 0.90. \end{aligned}$$

$$\therefore \quad P(L \mid M_1 \cap M_2) = 0.90 > P(L) = 0.10.$$

Case 2: $P(L \mid M_1 \cap M_2^c)$.

By conditional independence,

$$\begin{aligned} P(M_1 \cap M_2^c \mid L) &= P(M_1 \mid L)P(M_2^c \mid L) \\ &= 0.90(0.10) = 0.09, \\ P(M_1 \cap M_2^c \mid L^c) &= P(M_1 \mid L^c)P(M_2^c \mid L^c) \\ &= 0.10(0.90) = 0.09. \end{aligned}$$

$$\begin{aligned} P(M_1 \cap M_2^c) &= P(M_1 \cap M_2^c \mid L)P(L) \\ &\quad + P(M_1 \cap M_2^c \mid L^c)P(L^c) \\ &= 0.09(0.10) + 0.09(0.90) \\ &= 0.09. \end{aligned}$$

By Bayes' theorem,

$$\begin{aligned} P(L \mid M_1 \cap M_2^c) &= \frac{P(M_1 \cap M_2^c \mid L)P(L)}{P(M_1 \cap M_2^c)} \\ &= \frac{0.09(0.10)}{0.09} \\ &= 0.10. \end{aligned}$$

$$\therefore \quad P(L \mid M_1 \cap M_2^c) = P(L) = 0.10.$$

(c)

By the law of total probability,

$$\begin{aligned} P(M_1) &= 0.90(0.10) + 0.10(0.90) \\ &= 0.18, \\ P(M_2) &= 0.90(0.10) + 0.10(0.90) \\ &= 0.18. \end{aligned}$$

$$\therefore \quad P(M_1)P(M_2) = 0.18^2 = 0.0324.$$

From part (b),

$$P(M_1 \cap M_2) = 0.09.$$

$$\because \quad P(M_1 \cap M_2) = 0.09 \neq 0.0324 = P(M_1)P(M_2),$$

$$\therefore \quad M_1 \text{ and } M_2 \text{ are not independent.}$$

However, conditional independence gives

$$P(M_1 \cap M_2 \mid L) = P(M_1 \mid L)P(M_2 \mid L) = 0.81,$$

$$P(M_1 \cap M_2 \mid L^c) = P(M_1 \mid L^c)P(M_2 \mid L^c) = 0.01.$$

Moreover,

$$\begin{aligned} P(M_2 \mid M_1) &= \frac{P(M_1 \cap M_2)}{P(M_1)} \\ &= \frac{0.09}{0.18} = 0.50 \neq P(M_2) = 0.18. \end{aligned}$$

But

$$P(M_2 \mid M_1, L) = P(M_2 \mid L) = 0.90,$$

$$P(M_2 \mid M_1, L^c) = P(M_2 \mid L^c) = 0.10.$$

Thus, the programs are conditionally independent given the true email status, but not independent without conditioning, because both outputs depend on the same underlying email status.

Problem4

Consider a simplified model of an AI evaluation. In one run, an AI system answers two questions. Write C for correct and I for incorrect; the ordered outcomes CC, CI, IC, II also denote the corresponding events.

In parts (a) and (b), each answer is correct with probability $1/2$, independently of the other, so the four outcomes are equally likely.

An evaluator checks both answers against an answer key and may publish one of the following reports. Let M_C and M_I denote the events that the corresponding reports are published:

$$\begin{aligned} M_C: &\quad \text{“At least one answer is correct”}, \\ M_I: &\quad \text{“At least one answer is incorrect”}. \end{aligned}$$

At most one report is published per run, and every published report must be true.

(a) Initially, every run is reported. For CC, the evaluator publishes M_C; for II, the evaluator publishes M_I. For either CI or IC, a fair coin is used to choose between the two reports, each with probability $1/2$.

Using Bayes' rule, find

$$P(CC \mid M_C) \qquad \text{and} \qquad P(II \mid M_I).$$

(b) Replace the rule in part (a) by any randomized reporting rule based on the outcome. The evaluator may choose not to publish a report for any of the four outcomes. The two reports must each have positive probability and must satisfy

$$P(CC \mid M_C) \geq \frac{2}{3}, \qquad P(II \mid M_I) \geq \frac{3}{4}.$$

What is the smallest possible probability that no report is published? Specify a reporting rule for each of CC, CI, IC, II that attains your answer, and prove that no admissible rule can do better.

(c) Remove the independence assumption, but keep each answer's probability of being correct equal to $1/2$. The evaluator may choose a new randomized reporting rule, but must publish exactly one truthful report per run, with both report types having positive probability.

Let $t = P(CC)$. Determine all values of t for which such a rule satisfies both posterior inequalities in part (b).

Prove necessity and sufficiency, giving a valid rule for every claimed value of t. The rule may depend on t.

Solution

(a)

$$P(CC) = P(CI) = P(IC) = P(II) = \frac{1}{4}.$$

$$\begin{aligned} P(M_C \mid CC) &= 1, \\ P(M_C \mid CI) &= \frac{1}{2}, \\ P(M_C \mid IC) &= \frac{1}{2}, \\ P(M_C \mid II) &= 0. \end{aligned}$$

By the law of total probability,

$$\begin{aligned} P(M_C) &= \sum_{\omega \in \{CC,CI,IC,II\}} P(M_C \mid \omega) P(\omega) \\ &= 1 \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} \\ &= \frac{1}{2}. \end{aligned}$$

By Bayes' rule,

$$\begin{aligned} P(CC \mid M_C) &= \frac{P(M_C \mid CC) P(CC)}{P(M_C)} \\ &= \frac{1 \cdot \frac{1}{4}}{\frac{1}{2}} \\ &= \frac{1}{2}. \end{aligned}$$

Similarly,

$$\begin{aligned} P(M_I) &= 0 \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} \\ &= \frac{1}{2}. \end{aligned}$$

$$\begin{aligned} P(II \mid M_I) &= \frac{P(M_I \mid II)P(II)}{P(M_I)} \\ &= \frac{1 \cdot \frac{1}{4}}{\frac{1}{2}} \\ &= \frac{1}{2}. \end{aligned}$$

$$\therefore \quad P(CC \mid M_C) = P(II \mid M_I) = \frac{1}{2}.$$

(b)

Let N denote the event that no report is published.

$$P(CC \mid M_C) \geq \frac{2}{3}$$

$$\implies \frac{P(CC \cap M_C)}{P(M_C)} \geq \frac{2}{3}$$

$$\implies P(M_C) \leq \frac{3}{2} P(CC \cap M_C).$$

$$\because \quad P(CC \cap M_C) \leq P(CC) = \frac{1}{4}$$

$$\therefore \quad P(M_C) \leq \frac{3}{2} \cdot \frac{1}{4} = \frac{3}{8}.$$

Similarly,

$$P(II \mid M_I) \geq \frac{3}{4}$$

$$\implies P(M_I) \leq \frac{4}{3} P(II \cap M_I).$$

$$\because \quad P(II \cap M_I) \leq P(II) = \frac{1}{4}$$

$$\therefore \quad P(M_I) \leq \frac{4}{3} \cdot \frac{1}{4} = \frac{1}{3}.$$

$$\because \quad M_C \cap M_I = \varnothing$$

$$\begin{aligned} P(N) &= 1 - P(M_C) - P(M_I) \\ &\geq 1 - \frac{3}{8} - \frac{1}{3} \\ &= \frac{7}{24}. \end{aligned}$$

Achievability:

Consider the following reporting rule:

ω	$P(M_C \mid \omega)$	$P(M_I \mid \omega)$	$P(N \mid \omega)$
CC	1	0	0
CI	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{7}{12}$
IC	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{7}{12}$
II	0	1	0

$$\begin{aligned} P(M_C) &= \frac{1}{4}\left(1 + \frac{1}{4} + \frac{1}{4}\right) \\ &= \frac{3}{8}, \end{aligned}$$

$$\begin{aligned} P(M_I) &= \frac{1}{4}\left(\frac{1}{6} + \frac{1}{6} + 1\right) \\ &= \frac{1}{3}. \end{aligned}$$

$$P(CC \mid M_C) = \frac{P(CC \cap M_C)}{P(M_C)} = \frac{\frac{1}{4}}{\frac{3}{8}} = \frac{2}{3}.$$

$$P(II \mid M_I) = \frac{P(II \cap M_I)}{P(M_I)} = \frac{\frac{1}{4}}{\frac{1}{3}} = \frac{3}{4}.$$

$$\begin{aligned} P(N) &= 1 - P(M_C) - P(M_I) \\ &= 1 - \frac{3}{8} - \frac{1}{3} \\ &= \frac{7}{24}. \end{aligned}$$

$$\therefore \quad \min P(N) = \frac{7}{24}.$$

(c)

Let

$$t = P(CC).$$

$$P(C_1) = P(C_2) = \frac{1}{2},$$

where C_1, C_2 denote correctness of the first and second answers.

$$P(CI) = P(C_1) - P(CC) = \frac{1}{2} - t,$$
$$P(IC) = P(C_2) - P(CC) = \frac{1}{2} - t.$$

$$\begin{aligned} P(II) &= 1 - P(CC) - P(CI) - P(IC) \\ &= 1 - t - 2\left(\frac{1}{2} - t\right) \\ &= t. \end{aligned}$$

Thus,

$$\begin{aligned} P(CC) &= t, \\ P(CI) &= \frac{1}{2} - t, \\ P(IC) &= \frac{1}{2} - t, \\ P(II) &= t, \end{aligned} \qquad 0 \le t \le \frac{1}{2}.$$

Necessity:

Let

$$p = P(M_C).$$

Exactly one truthful report is published per run:

$$P(M_I) = 1 - p, \qquad 0 < p < 1.$$

$$P(M_C \mid CC) = 1, \qquad P(M_I \mid II) = 1.$$

$$\implies P(CC \cap M_C) = t, \qquad P(II \cap M_I) = t.$$

The first posterior constraint gives

$$P(CC \mid M_C) = \frac{t}{p} \ge \frac{2}{3}$$

$$\implies \quad p \le \frac{3}{2}t.$$

The second posterior constraint gives

$$P(II \mid M_I) = \frac{t}{1-p} \ge \frac{3}{4}$$

$$\implies \quad 1 - p \le \frac{4}{3}t$$

$$\implies \quad p \ge 1 - \frac{4}{3}t.$$

$$\therefore \quad 1 - \frac{4}{3}t \le p \le \frac{3}{2}t.$$

$$\implies 1 - \frac{4}{3}t \leq \frac{3}{2}t$$

$$\implies 1 \leq \frac{17}{6}t$$

$$\implies t \geq \frac{6}{17}.$$

$$\because \quad t \leq \frac{1}{2}$$

$$\therefore \quad \frac{6}{17} \leq t \leq \frac{1}{2}.$$

Sufficiency:

For every

$$t \in \left[\frac{6}{17}, \frac{1}{2}\right],$$

define

$$q(t) = \begin{cases} \dfrac{t}{2(1-2t)}, & \dfrac{6}{17} \leq t \leq \dfrac{2}{5}, \\ 1, & \dfrac{2}{5} < t \leq \dfrac{1}{2}. \end{cases}$$

Then

$$0 \leq q(t) \leq 1.$$

Consider the reporting rule:

ω	$P(M_C \mid \omega)$	$P(M_I \mid \omega)$
CC	1	0
CI	$q(t)$	$1 - q(t)$
IC	$q(t)$	$1 - q(t)$
II	0	1

Each row sums to 1, and every report is truthful.

Case 1:

$$\frac{6}{17} \leq t \leq \frac{2}{5}.$$

$$\begin{aligned} P(M_C) &= t + 2\left(\frac{1}{2} - t\right) q(t) \\ &= t + (1-2t)\frac{t}{2(1-2t)} \\ &= \frac{3}{2}t. \end{aligned}$$

$$P(M_I) = 1 - \frac{3}{2}t.$$

$$P(CC \mid M_C) = \frac{t}{\frac{3}{2}t} = \frac{2}{3}.$$

$$P(II \mid M_I) = \frac{t}{1 - \frac{3}{2}t}.$$

$$\because \quad t \geq \frac{6}{17}$$

$$\implies \frac{17}{2}t \geq 3$$

$$\implies 4t \geq 3 - \frac{9}{2}t$$

$$\implies \frac{t}{1 - \frac{3}{2}t} \geq \frac{3}{4}.$$

$$\therefore \quad P(CC \mid M_C) \geq \frac{2}{3}, \qquad P(II \mid M_I) \geq \frac{3}{4}.$$

Case 2:

$$\frac{2}{5} < t \leq \frac{1}{2}.$$

$$q(t) = 1.$$

$$\begin{aligned} P(M_C) &= P(CC) + P(CI) + P(IC) \\ &= t + 2\left(\frac{1}{2} - t\right) \\ &= 1 - t. \end{aligned}$$

$$P(M_I) = P(II) = t.$$

$$P(CC \mid M_C) = \frac{t}{1 - t}.$$

$$\because \quad t > \frac{2}{5}$$

$$\implies 5t > 2$$

$$\implies 3t > 2(1 - t)$$

$$\implies \frac{t}{1 - t} > \frac{2}{3}.$$

Furthermore,

$$P(II \mid M_I) = \frac{t}{t} = 1 \geq \frac{3}{4}.$$

$$\therefore \quad P(CC \mid M_C) \geq \frac{2}{3}, \qquad P(II \mid M_I) \geq \frac{3}{4}.$$

In both cases,

$$P(M_C) > 0, \qquad P(M_I) > 0.$$

Combining necessity and sufficiency,

$$\therefore \quad t \in \left[\frac{6}{17}, \frac{1}{2}\right].$$